

What the standards say, and what the simulator can see

2,463 candidate rules extracted from 16 grid codes and reliability standards, measured against what a Grid2Op observation can actually evaluate. Every figure below is counted from artifacts on disk, not estimated.

1. The corpus by subject, and whether Grid2Op can express it

Grouped by what each rule is physically about; every candidate appears exactly once. The last three columns are its actual fate in the pipeline.

Subject family	n	% of corpus	Time-bound	Grid2Op observable	Translated	Guarded	Validated	Usable for N-1 screening
Frequency / RoCoF / governor droop	655	26.6%	117	No - not modelled at any tier	0	0	0	No. Needs a dynamic simulator.
Voltage magnitude & bands	560	22.7%	129	Yes - v_or with per-line base kV	43	21	1	Partial. Expressible, but the grid-wide band is vacuous on 2 of 3 grids and never vetoes.
Protection & equipment settings	305	12.4%	39	No - no relays, no winding temperatures	0	0	0	No.
Timing / duration only	218	8.9%	214	Partial - step counts only (timestep_overflow)	0	0	0	No as written. A subset is recoverable as step-count rules.
Thermal loading & current	142	5.8%	12	Yes - rho, a_or, thermal limit	9	9	9	Yes. This family is the entire working shield.
Reactive power & power factor	133	5.4%	11	Yes - q_or, gen_q	4	0	0	Tried, failed. Predicates measured inverted.
Miscellaneous boolean / equipment-state	131	5.3%	0	No - mostly non-physical predicates	0	0	0	No.
Active power, dispatch & capacity	90	3.7%	12	Yes - p_or, gen_p, gen_margin_up/down, pmin/pmax	1	1	1	Recoverable. Variables exist; the vocabulary has no name for them.
Load shedding & system balance	81	3.3%	6	Partial - gen_p / load_p, detachment flags	1	1	0	Marginal. The one rule that got through is inert.
Administrative / process / reporting	72	2.9%	0	No - not a physical check	0	0	0	No.
Topology, connectivity & switching	51	2.1%	6	Yes - topo_vect, connectivity, cooldowns	0	0	0	Recoverable and untouched. Fully observable, zero coverage.
Power quality (flicker, harmonics)	25	1.0%	1	No - steady-state solver only	0	0	0	No.
TOTAL	2,463	100%	547		58	32	11	

Read it this way: the two largest families — frequency and protection, 39% of the corpus between them — are unreachable at any tier, and no prompt or vocabulary change touches them. Thermal loading is 5.8% of the corpus and supplies 9 of the 11 validated rules; counting R_2165 (*power_flow* > *technical_limit*, grouped under active power but translated to *loading_pct* > 100) it supplies 10 of 11. The eleventh is the voltage affirmation. Assignment is first-match-wins on the condition's own variables, falling back to the explanation text; the three fate columns reconcile with the stage totals (58 translated, 32 kept by the polarity guard, 11 validated).

2. What a Grid2Op snapshot can check, and what the corpus supplied

The left column is the observable families available from a Grid2Op observation, its static grid constants, and one *simulate()* call. The right columns are what the knowledge graph actually holds against each: 4 served rules, 58 explanatory rules, 26 predicates, 5 variables.

Grid2Op observable family	Rules in KG	Channel	Can veto?	Status
Line loading (rho)	10	BLOCK	Yes	The only veto path in the system. 3 served rules, 10 clauses, 4 documents.
Line voltage (v_or, per-unit band)	39	WARN / NORMAL	No	Largest expressible family; contributes nothing to any reported metric.
Power factor (p_or, q_or)	4	NOT_APPLICABLE	No	Demoted on measurement - the predicates fire more often when risk is lower.
Generation / load balance (gen_p, load_p)	1	INERT	No	Never fires on any grid.
Line current, active / reactive / apparent power	0	-	-	Observable, no rule.
Line status, tripped-line counts	0	-	-	Observable, no rule.
Topology, bus splitting, islanding	0	-	-	Observable, no rule.
Overload duration, cooldowns, maintenance	0	-	-	In a live observation, not stored, no rule.
Dispatch, headroom, ramps, curtailment, storage	0	-	-	In a live observation, not stored, no rule.
Static ratings (thermal limit, pmin/pmax, costs)	0	-	-	Available as grid constants, no rule.
Post-contingency state (simulate)	0	-	-	The label the model is scored against. Hand-written in validate_n1, not extracted.

The result in four lines

- **5 of 14** vocabulary variables appear in the knowledge graph at all; **26 predicates collapse to 4 physical checks**, of which two do any work.
- **One check can stop a prediction** — *loading_pct* > 100 — served as three deduplicated rules and corroborated by ten clauses across four documents.
- **Voltage is the near-miss.** It is the most-stated family in the corpus and contributes nothing measurable: the band is breached on 100% of frames on two of three grids, both before and after the contingency.
- **Two families are recoverable without new data** — topology/switching and dispatch headroom. Both are blocked by the 14-variable vocabulary, not by the simulator.

Sources: rules_35b/ (stage 1), translated_rules/ (stage 2 and the polarity guard), validated_translated/ (stage 3), shield_corpus/all_rules_channels.jsonl, kg/knowledge_graph.json. Table 1 is produced by evaluation/subject_families.py; the stage partition it reconciles against is verified by evaluation/corpus_accounting.py.